Mahavir Education Trust's

SHAH & ANCHOR KUTCHHI ENGINEERING COLLEGE

An Autonomous Institute Affiliated to University of Mumbai • Approved by AICTE, New Delhi & Govt. of Maharashtra

NAAC Accredited 'A' Grade (3.16 CGPA) • NBA Accredited UG Programs (Comp. Engg., IT, EXTC) • ISO 9001:2015 • DTE Institute Code: 3148

Mahavir Education Trust's Chowk, W. T. Patil Marg, Next to Duke's Co., Chembur, Mumbai – 400 088 • Tel: 022-25580854 • https://www.sakec.ac.in/

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ACADEMIC YEAR: 2025-26 • SEMESTER: V • SCHEME: NEP 2020 / AICTE Model Curriculum • TYPE: Integrated (Theory + Practical)

Course Name:	**Cloud-Native Distributed Systems and Microservices**		
Course Code:	**22CS501**	**Course Type:**	Integrated (Theory + Practical)
Credits (L-T-P-C):	3 - 0 - 2 - **4**	**Total Contact Hours:**	**45 Hours**
Theory Hours:	45 Hours	**Practical/Lab Hours:**	30 Hours
Self-Study / Project:	15 Hours	**Total Marks:**	**170 Marks**

1. Course Outcomes (COs) & Multi-Domain Bloom's Taxonomy

CO ID	Course Outcome Statement	Domain	Taxonomy Level	Target %
CO1	Explain the theoretical principles and protocol state machines governing the technology domain. for Cloud-Native Distributed Systems and Microservices.	**Cognitive**	L2: Understand	70.0%
CO2	Configure, instrument, and operate development toolchains, container runtimes, and local clusters according to technical specifications. for Cloud-Native Distributed Systems and Microservices.	**Psychomotor**	P2: Manipulation	70.0%
CO3	Diagnose system failures, benchmark operational bottlenecks, and profile latency profiles under stress. for Cloud-Native Distributed Systems and Microservices.	**Cognitive**	L4: Analyze	70.0%
CO4	Implement and execute end-to-end automated pipelines, real-time streaming topologies, and API contracts with measurable accuracy. for Cloud-Native Distributed Systems and Microservices.	**Psychomotor**	P3: Precision	70.0%
CO5	Adhere to professional engineering standards, open-source compliance, ethical zero-trust postures, and collaborative peer code review. for Cloud-Native Distributed Systems and Microservices.	**Affective**	A3: Valuing	70.0%
CO6	Synthesize a functional capstone engineering solution incorporating hardware acceleration or production-grade cloud architectures. for Cloud-Native Distributed Systems and Microservices.	**Cognitive**	L6: Create	70.0%

Domain Standards: Cognitive (L1: Remember to L6: Create), Psychomotor (P1: Imitation to P5: Naturalization for Lab/Hands-on), Affective (A1: Receiving to A5: Characterization for Team/Ethics).

2. Program Specific Outcomes (PSOs)

- **PSO1**: Professional Skills: Design and develop enterprise cloud and edge systems.
- **PSO2**: Industrial Competence: Implement streaming data architectures and real-time AI pipelines.
- **PSO3**: Research & Ethics: Formulate secure, sustainable computing solutions.

3. CO-PO and CO-PSO Articulation Matrix (Course Articulation)

CO ID	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	2	2	2	2
CO2	3	2	-	3	3	-	-	-	-	-	-	2	2	2	2
CO3	3	3	2	3	3	-	-	-	-	-	-	2	3	3	3
CO4	3	3	2	3	3	-	-	-	-	-	-	2	3	3	3
CO5	3	3	3	-	3	-	-	3	2	2	-	2	3	3	3
CO6	3	3	3	-	3	-	-	-	2	2	-	2	3	3	3
Average	**3.0**	**2.8**	**2.5**	**3.0**	**3.0**	**-**	**-**	**3.0**	**2.0**	**2.0**	**-**	**2.0**	**2.7**	**2.7**	**2.7**

Correlation Levels: 3: Substantial (High), 2: Moderate (Medium), 1: Slight (Low), -: No Correlation.

4. Detailed Unit / Module-Wise Syllabus

Module 1: Foundational Concepts and Theoretical Framework

Contact: 9 Hours **Weightage: 20.0%**

Cognitive Focus: L1-L2

- Introduction to Cloud-Native Distributed Systems and Microservices and core domain principles
- Formal models, architectural taxonomies, and invariant properties
- Mathematical foundations, complexity bounds, and state representation
- Design space trade-offs, consistency criteria, and boundary conditions

Module 2: Core Algorithms and Design Patterns

Contact: 9 Hours **Weightage: 20.0%**

Cognitive Focus: L2-L3

- Fundamental algorithms, control flows, and scheduling mechanisms
- Component interactions, interface contracts, and encapsulation
- State synchronization, resource management, and partition recovery
- Analytical verification and baseline validation benchmarks

Module 3: System Architecture and Optimization

Contact: 9 Hours **Weightage: 20.0%**

Cognitive Focus: L3-L4

- Modular decomposition, layering, and concurrency abstractions
- High-performance caching, replication, and data partitioning
- Fault-tolerant consensus, failure detectors, and self-healing topologies
- Empirical profiling, bottleneck detection, and latency mitigation

5. Assessment Scheme and Examination Pattern

Module 4: Modern Engineering Workflows & Edge AI	Contact: 9 Hours	Weightage: 20.0%

Cognitive Focus: L4-L5 **Industry Integration:** Edge AI, vLLM

- Integration of Edge AI into modern operational pipelines
- Asynchronous event processing using vLLM and message brokers
- Distributed telemetry, observability, and structured logging
- Zero-trust security, containerized deployment, and CI/CD pipelines

Module 5: Advanced Industry Applications & TensorRT-LLM	Contact: 9 Hours	Weightage: 20.0%

Cognitive Focus: L5-L6 **Industry Integration:** TensorRT-LLM, Kafka Streams

- Edge computing architectures, TinyML, and hardware acceleration with TensorRT-LLM
- High-throughput LLM serving and model optimization utilizing Kafka Streams
- Fault injection testing, chaos engineering, and resilience metrics
- Scalability benchmarking and compliance with modern enterprise standards

Evaluation Component	Max Marks	Assessment Mechanism & Rubrics
Continuous Internal Assessment (CIA)	**20 Marks**	Quizzes, Surprise Tests, Open-book Case Analysis & Active Learning.
Mid-Semester Examination (MSE)	**30 Marks**	1.5-hour centralized written test covering foundational Modules 1 to 3.
Practical Internal Assessment (CIAP)	**25 Marks**	Continuous evaluation of lab experiments, code reviews, and logbooks.
End-Semester Examination (ESE - Theory)	**50 Marks**	3-hour comprehensive written evaluation across all modules.
End-Semester Practical Exam (ESEP)	**25 Marks**	Hands-on lab implementation, viva-voce, and runtime verification.
Self-Learning Assessment (SLA)	**20 Marks**	NPTEL/MOOC certification, capstone mini-project, or industrial seminar.
Total Aggregated Course Marks	**170 Marks**	**Passing Threshold: 40.0% Aggregated**

6. Textbooks and Reference Books (IEEE Academic Format)

Textbooks:

1. Martin Kleppmann, *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, 1st Edition, O'Reilly Media, 2023. [ISBN: 978-1449373320]
2. Andrew S. Tanenbaum and Maarten van Steen, *Distributed Systems: Principles and Paradigms*, 4th Edition, Pearson Education, 2024. [ISBN: 978-1543057386]

Reference Books & Web Resources:

1. Brendan Burns, *Designing Distributed Systems: Patterns and Paradigms for Scalable, Reliable Services*, 2nd Edition, O'Reilly Media, 2024. [ISBN: 978-1492031772]
2. IEEE / ACM Joint Curriculum Committee, *Computer Science Curricula 2023: Guidelines for Associate and Baccalaureate Programs in Computing*, Final Edition, IEEE-CS & ACM Press, 2024.

*System Platform Engineered & Developed by **Dr. Rohan Borgalli**, Shah & Anchor Kutchhi Engineering College, Mumbai (https://www.sakec.ac.in/).*

Course Coordinator (Faculty)	Head of Department (HOD)	Dr. Bhavesh Patel (Principal)
Prepared by Faculty In-Charge	*Dept. of Computer Science & Engineering, SAKEC*	*Shah & Anchor Kutchhi Engg. College*